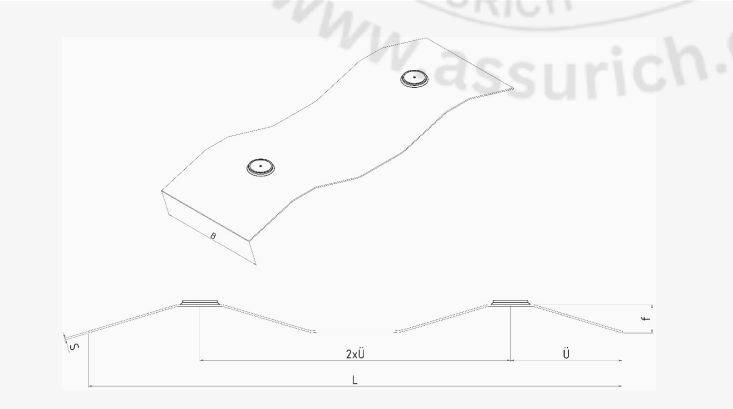
Panel handling
Technical configuration

Panel overhang

The correct configuration of a vacuum lifter is essential to ensure adequate safety. In particular, the permissible overhang—especially for thin-walled workpieces—must be taken into account. If the overhang is too large, the material may bend undesirably, which in turn can place excessive load on individual suction grippers.



1 L = Panel length

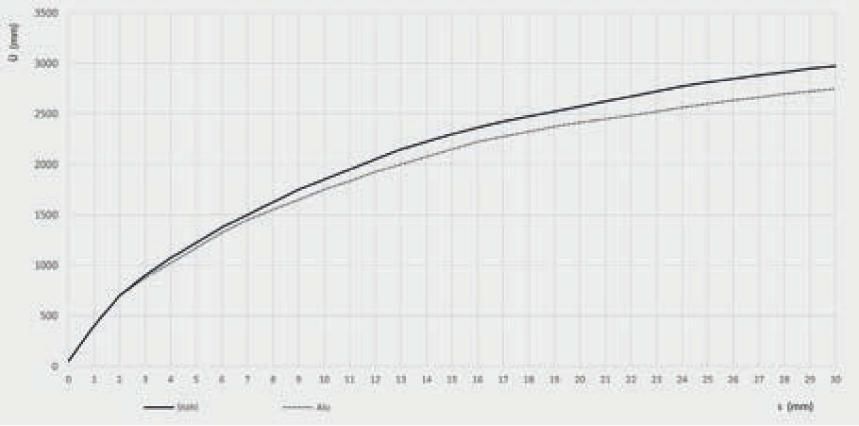
2 B =Panel width

3 Ü = Permissible overhang

4 f = Deflection

5 S = Sheet thickness

Overhang and deflection of panel material during suction



Quick selection table for permissible overhangs

Panel handling
Technical configuration

Overhang table

In some cases, the exact deflection of the panel material must be known, for example to clear interfering edges or machine components. The following table provides approximate deflection values for steel, aluminum, wood, and glass panels, depending on panel thickness and overhang.

Deflection f depending on sheet thickness and sheet overhang

Overhang (mm) / sheet thickness (mm)	100	200	300	400	500	600	800	1.000	1.250	1.500	2.000	2.500	3.000
0,5	5	25	75	250	500								
1			5	10	20	50	200	500					
2					5	10	50	150	400				
3						5	20	75	200	500			
4							10	40	100	250			
6							5	20	45	125	500		
8								10	30	75	300		
10								5	20	50	200	500	
12									10	30	125	400	
15									7,5	20	75	250	
20									5	10	50	150	350
25										5	30	75	250
30										2,5	20	40	150

Number of suction grippers

Based on the overhang values, the required number of suction grippers can be estimated as follows:

Number along the length $n_L = L / (2xÜ)$

Number along the width $n_B = B / (2xÜ)$

Total number $n_G = n_L \times n_B$